

Poisson Distribution



Conditions for a Poisson distribution

- 1 State the key conditions required for a random variable to be modeled by a Poisson distribution.
- 2 Explain why the number of texts a person receives in a randomly chosen hour might reasonably be modeled by a Poisson distribution, but the number of texts received right around a pre-arranged deadline might not.

Computing Poisson probabilities

- 3 A shop receives an average of 3 customers per minute, modeled by $X \sim \text{Po}(3)$. Find $P(X = 5)$, to three decimal places.
- 4 Using the same shop, find $P(X = 1)$, to three decimal places.
- 5 Using the same shop, find $P(X \leq 2)$.

Mean and variance of a Poisson distribution

- 6 State the mean and variance of $X \sim \text{Po}(\lambda)$, and find the standard deviation when $\lambda = 9$.
- 7 A random variable $X \sim \text{Po}(7)$. State $E(X)$, $\text{Var}(X)$, and the standard deviation, to three decimal places.

Sums of independent Poisson variables

- 8 Two independent Poisson processes have rates $\lambda_1 = 4$ and $\lambda_2 = 6$ per hour. State the distribution of the total number of events, and find its mean and variance.
- 9 Three independent Poisson processes have rates $\lambda_1 = 2$, $\lambda_2 = 4$, $\lambda_3 = 5$ per hour. Find the mean and variance of the total number of events across all three processes in one hour.

Poisson as an approximation to the binomial

- 10 Explain the conditions under which a binomial distribution $X \sim B(n, p)$ can be well-approximated by a Poisson distribution, and state the value of λ to use.

- 11 A factory produces 2000 items, each independently defective with probability 0.0025. Using the Poisson approximation, find the probability that exactly 2 items are defective, to three decimal places.
- 12 A survey mails 5000 questionnaires, each independently containing a printing error with probability 0.0006. Using the Poisson approximation, find the probability that no questionnaires contain an error, to four decimal places.
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Using cumulative Poisson probabilities

- 13 A call center receives calls at $\lambda = 6$ per 10-minute interval. Find the probability of receiving more than 3 calls, given $P(X = 0) \approx 0.00248$, $P(X = 1) \approx 0.01487$, $P(X = 2) \approx 0.04462$, $P(X = 3) \approx 0.08924$.
- 14 A website receives visitors at $\lambda = 4$ per minute. Find $P(1 \leq X \leq 3)$, given $P(X = 0) \approx 0.0183$, $P(X = 1) \approx 0.0733$, $P(X = 2) \approx 0.1465$, $P(X = 3) \approx 0.1954$.
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Finding an unknown parameter

- 15 A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 0) = 0.0498$. Find λ , to the nearest whole number.
- 16 A random variable $X \sim \text{Po}(\lambda)$ has standard deviation 2. Find λ , and then find $P(X = \lambda)$, to three decimal places.
- 17 A random variable $X \sim \text{Po}(\lambda)$ satisfies $P(X = 1) = 0.1494$, where λ is a whole number less than 5. Using $P(X = 1) = \lambda e^{-\lambda}$, determine λ by trial.
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Poisson processes over different intervals and conditional probability

- 18 Machine faults occur at an average rate of $\lambda = 2$ per day. Find the probability that exactly 2 faults occur today, given that at least one fault occurs today, to three decimal places, using $P(X = 0) \approx 0.1353$ and $P(X = 2) \approx 0.2707$.
- 19 A radioactive source emits particles at a rate of $\lambda = 1.2$ per second, modeled by a Poisson distribution. Find $P(X = 2)$ for a 3-second interval, to three decimal places, using the fact that the parameter scales with the length of the interval.
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A well-designed word problem: quality control at a bottling plant

- 20** This question has four parts. Faulty caps occur at random in bottling cases with an average rate of $\lambda = 1.5$ per case. (a) Find $P(X = 0)$ and $P(X = 1)$ for a randomly selected case, to four decimal places. (b) Find the probability that a case has at least 2 faulty caps. (c) The plant inspects 40 cases per day. Using the fact that a sum of independent Poisson variables is itself Poisson, state the distribution of the total number of faulty caps found across all 40 cases in a day, and find its mean. (d) Since this total has a large λ , it can be approximated by $N(\lambda, \lambda)$. Using a continuity correction and $\Phi(1.36) \approx 0.9131$, estimate the probability that more than 70 faulty caps are found in a day.